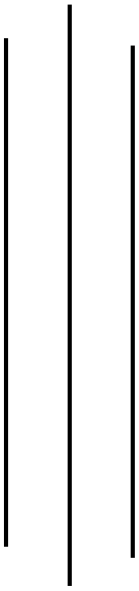
Lumbini Technological University
Institute of Engineering and Information Technology

Banke, Lumbini Province

Syllabus



B.Tech in CS & AI

Year/Semester: II/I

Statistics

Course Code	BAI201	**Year/Semester**	II/I
Credit Weightage	3	**Lecture**	3 hrs/wk
		Tutorial	1 hrs/wk
		Practical	2 hrs/wk
		Total	96 hrs

1. Course Description

This course introduces B. Tech students to the fundamental principles of statistics and its applications in engineering and scientific fields. Students will learn to analyze and interpret data, make informed decisions based on statistical evidence, and apply statistical techniques to solve real-world engineering problems. The course aims to develop students' understanding of statistical concepts and their ability to use statistical tools effectively.

2. Course Objectives

By the end of the course, students should be able to:

- Understand the basic concepts of statistics and its relevance in engineering and scientific research.
- Apply descriptive statistics to summarize and present data effectively.
- Comprehend probability theory and its role in modeling uncertain events.
- Analyze and interpret various types of probability distributions, including discrete and continuous distributions.
- Use sampling distributions to make inferences about population parameters.
- Conduct hypothesis tests to make decisions based on sample data.
- Apply regression analysis to model relationships between variables and make predictions.
- Design and analyze experiments to investigate cause-and-effect relationships.
- Apply statistical software to analyze data, perform calculations, and create visualizations.

3. Course Details

3.1 Theory (48 hrs)

Unit 1: Descriptive Statistics (10 hrs)

Introduction to Statistics- descriptive statistics and inferential statistics; types of Data-Qualitative vs. Quantitative data, levels of measurements- nominal level data, ordinal level data, interval level data, and ratio level data and variables.

Presentations of data: Frequency Distribution-Simple frequency distribution, relative frequency distribution, and percentage frequency distribution; Rule of constructing the frequency distribution, data array, stem and leaf display; Graphical presentations; bar char and pie chart, histogram, cumulative frequency distribution and Ogive. Descriptive measure of data: Quartiles, Deciles, and Percentiles; five number summary, box-and-Whisker plot, shape of the data-skewness and kurtosis; measure of central tendency-Mean, Median, and Mode; Measures of dispersion: Range, Average Deviation Measures, Standard Deviation, Variance, Relative Dispersion and Coefficient of Variation.

Students also practice and use the statistical software such as SPSS, STATA, R, or MS Excel etc. to calculate mean, standard deveins, skewness, kurtosis, and to visualize the data in graphical form.

Unit 2: Basic Probability (5 hrs)

Introduction to basic probability theory, Approaches to assigning probabilities- classical probability, empirical probability and subjective probability; Terminology in probability, principles of counting- multiplication, permutation and combination; contingency table; Probabilities Rules- Additional and multiplication; Probabilities under conditions of Statistical Independence, Probabilities under conditions of statistical dependence; Bayes' theorem.

Unit 3: Random Variables and mathematical expectation (3 hrs)

Random Variables- discrete random variable and continuous random variable, mathematical expectation of random variables, additional and multiplicative rule of mathematical expectations.

Unit 4: Probability Distributions (9 hrs)

Probability distribution function, joint probability function of two variables; Discrete probability distribution- Binomial Probability Distribution, Poisson Probability Distribution; Continuous probability distribution- Normal probability distributions, standard normal probability distribution function, normal distribution as an approximation of Binomial and Poisson probability distribution.

Students practice and use the statistical software such as SPSS, STATA, R, or MS Excel etc. to calculate the probabilities

Unit 5: Sampling Distributions and Estimation (7 hrs)

Introduction to Sampling, reasons for sampling, defining the populations, statistics vs. parameters, sampling methods- probability sampling and non-probability sampling methods; Sampling Distributions of the sample mean and proportions, concept of the Central Limit Theorem, weak law of large number,

Estimations: Introduction of basic terms of inferential statistics such as populations, sample, parameters, significance level, confidence level, critical value, degree of freedom; standard error; method of estimations- point estimations and interval estimations for mean and proportions, determining the sample size for the mean and proportion. Relations between sample size and margin of error.

Students practice and use the statistical software such as SPSS, STATA, R, or MS Excel etc. to calculate the confidence interval and sample size

Unit 6: Hypothesis Testing (10 hrs)

Introduction to hypothesis testing, Null and alternative hypotheses, Type I and Type II errors; t-tests for single mean and difference to two mean, z-tests for single mean and difference to two mean, paired t-test, Chi-Square test of goodness of fit, chi-square test for independence to two attributes. One way analysis of variance for difference of more than two means.

Students practice and use the statistical software such as SPSS, STATA, R, or MS Excel etc. to run hypothesis test the interpret the findings generated from the software to make informed decision.

Unit 7: Simple correlation and regression analysis (4 hrs)

Introduction, correlation analysis, regression equation using least squares method, measures of variation - computing the sum of squares; the coefficient of determination, standard error of the

estimate; Inference about the population slope, correlation coefficients, and regression equation using t-test and F-test; confidence interval for the slope, estimation of mean values; Limitations and errors.

Students also practice and use the statistical software such as SPSS, STATA, R, or MS Excel etc. to run correlation and regression analysis and interpret the findings generated from the software to make informed decision.

3.2 Tutorials

Needs tutorial class for practical session for the use of software to analyse the data in unit 1, 5, 6 and 7.

3.3 Laboratory Work

In the practical work, students should be able to use the statistical software such as MS Excel, SPSS, STATA or R to calculate and run the statistical tools such as mean, standard deviations, correlations, regression, hypotheses etc. of the dataset.

4. Evaluation Scheme

Evaluation of student's performances is divided into two main parts: (1) Internal Evaluation and (2) Semester examination, with breakdowns as follows:

Evaluation	**Components**	**Weightage**	**Full Marks**	**Pass Marks**
Internal Evaluation	Class Attendance and Performance	5	20	10
	Assignment	10		
	Seminar/Project/Presentation	5		
	Practical/Lab Examination	20	20	10
	Internal Examination	**20**	**20**	**10**
	Total Internal Marks		**60**	**30**
Semester Examination			**40**	**20**

5. Books

- Ronald E. Walpole, Raymond H. Myers, Sharon L. Myers and Keying Ye (2012). Probability and Statistics for Engineers & Scientist. 9th Ed. Printice Hall.
- Michael Baron (2013). Probability and Statistics for Computer Scientists. 2nd Ed., CRC Press, Taylor & Francis Group, A Chapman & Hall Book.
- Douglas C. Montgomery & George C. Runger (2003). Applied Statistics and Probability for Engineers. 3rd Ed. John Willey and Sons, Inc.
- Richard A. Johnson (2001). Probability and Statistics for Engineers. 6th Ed. Pearson Education, India
- Jay L. Devore (2012). Probability and Statistics for Engineering and the Sciences. 8th Ed. Cengage Learning, California Polytechnic State University, San Luis Obispo.

Database Management System

Course Code	BAI202	**Year/Semester**	II/I
Credit Weightage	3	**Lecture**	3 hrs/wk
		Tutorial	1 hrs/wk
		Practical	2 hrs/wk
		Total	96 hrs

1. Course Description

This course is designed to provide students with a comprehensive understanding of database management system concepts. The course covers different aspects of database management including, basics of database, data modelling using ER diagram, relational data model and relational algebra, structured query language, normalization, transaction processing, concurrency control and database recovery techniques. Through a combination of lectures, practical exercises, and projects, students will gain the necessary skills to design and develop databases.

2. Course Objectives

The course aims to achieve the following objectives:

- Introduce different concepts of database management systems.
- Provide concepts and skills of designing logical data models
- Familiarize with relational algebra
- Provide concepts of structured query language
- Present concepts of normalization
- Acquaint with database transactions
- Deliver concepts of concurrency control and database recovery techniques

3. Course Details

3.1 Theory (48 Hrs)

Unit 1: Foundations of Database (5 Hrs.)

Introduction; Characteristics of the Database; DBMS; Actors and Workers in DBMS; Advantages of DBMS; Data Models; Schema and Instance; Three Schema Architecture; Data Abstraction and Independence; Database Language, Database System Environment

Unit 2: Entity-Relationship Model (7 Hrs.)

Entity Relationship Model; Entity Types, Entity Sets, Attributes, and Keys; Relationship Types, Relationship Sets, Mapping and Structural Constraints; Weak Entity Types; ER Diagram Notations; Naming Conventions in ER Diagram; ER-to-Relational Mapping; Extended ER Model; Specialization and Generalization; Constraints on Specialization and Generalization

Unit 3: The Relational Data Model and Relational Algebra (7 Hrs.)

Relational Model; Domains, Attributes, Tuples and Relations; Relational Model Constraints; Relational Database Schemas; Relational Algebra: Unary Relational Operations: SELECT, PROJECT, RENAME; Relational Algebra Operations from Set Theory; Binary Relational Operations: JOIN and DIVISION; Aggregate Functions; Outer Joins

Unit 4: SQL **(11 Hrs.)**

Data Types; SQL Constraints; DDL Statements; DML Statements: Select, Aliasing, Substring Comparison, Arithmetic Operators, Order By, Null Values, Nested Query, Correlated Nested Query, Exists, Set Comparison, Inner Join, Outer Join, Aggregate Functions, Group By Having; INSERT, DELETE, and UPDATE Statements; Views, Stored Procedure, Triggers

Unit 5: Normalization **(6 Hrs.)**

Design Guidelines for Relational Databases; Functional Dependencies; Normalization and its importance; First, Second and Third Normal Forms; Boyce-Codd Normal Form; Multivalued Dependency; Fourth Normal Form; Properties of Relational Decomposition

Unit 6: Concepts of Transaction Processing **(6 Hrs.)**

Introduction to Transaction Processing; Transaction and System Concepts; ACID Properties of Transactions; Characterizing Schedules Based on Recoverability; Characterizing Schedules Based on Serializability

Unit 7: Concurrency Control and Database Recovery **(6 Hrs)**

Concurrency Control and its purpose; Two-Phase Locking Technique for Concurrency Control; Concurrency Control Based on Timestamp Ordering; Concepts of Recovery; NO-UNDO/REDO Recovery Based on Deferred Update; Recovery Techniques Based on Immediate Update; Shadow Paging; The ARIES Algorithm; Database Backup and Recovery from Catastrophic Failures

3.2 List of Tutorials

Unit-specific exercises focused on problem-solving.

3.3 Laboratory Works

The practical work includes modelling ER and EER diagrams using appropriate CASE tools. The practical extends writing SQL query scripts to define and manipulate databases. Real world database projects should be designed and developed as a group project under the practical work.

4. Evaluation Scheme

Evaluation of student's performances is divided into two main parts: (1) Internal Evaluation and (2) Semester examination, with breakdowns as follows:

Evaluation	**Components**	**Weightage**	**Full Marks**	**Pass Marks**
Internal Evaluation	Class Attendance and Performance	5	20	10
	Assignment	10		
	Seminar/Project/Presentation	5		
	Practical/Lab Examination	20	20	10
	Internal Examination	**20**	**20**	**10**
	Total Internal Marks		**60**	**30**
Semester Examination			**40**	**20**

5. Books:

- Fundamentals of Database Systems; Seventh Edition; Ramez Elmasri, Shamkant B. Navathe; Pearson
- Database System Concepts; Seventh Edition; Avi Silberschatz, Henry F Korth, S Sudarshan; McGraw-Hill
- An introduction to Database Systems; Eighth Edition; CJ Date, Pearson
- Database Design and Relational Theory: Normal Forms and All That Jazz; Second Edition; CJ Date; Apress
- Database Management Systems; Third Edition; Raghu Ramakrishnan, Johannes Gehrke; McGraw-Hill
- A First Course in Database Systems; Third Edition; Jaffrey D. Ullman, Jennifer Widom; Pearson
- Learning SQL: Generate, Manipulate, and Retrieve Data; Third Edition; Alan Beaulieu; O'Reilly Media

Web Technology

Course Code	BAI203	**Year/Semester**	II/I
Credit Weightage	3	**Lecture**	3 hrs/wk
		Tutorial	1 hrs/wk
		Practical	2 hrs/wk
		Total	96 hrs

1. Course Description

This course is designed to provide students with a comprehensive understanding of web development and the underlying technologies that power the World Wide Web. The course covers various aspects of web technology, including HTML, CSS, JavaScript, server-side scripting, web servers, databases, and web application development. Through a combination of lectures, practical exercises, and projects, students will gain the necessary skills to create dynamic and interactive websites.

2. Course Objectives

The course aims to achieve the following objectives:

- Introduce students to the fundamental concepts and technologies that underpin the World Wide Web.
- Provide a solid foundation in HTML and CSS for web page design.
- Familiarize students with JavaScript and teach them how to use it to create dynamic and interactive elements on web pages.
- Introduce server-side scripting languages, such as PHP or Python, for web application development.
- Teach students the process of developing web applications using frameworks and libraries.
- Explore the integration of web applications with databases.
- Provide students with knowledge of web server configuration and deployment techniques.
- Educate students about common security vulnerabilities in web applications and teach them techniques to secure web applications against threats.
- Foster the ability to work collaboratively in a team environment, enabling students to effectively contribute to web development projects, communicate ideas, and coordinate tasks.
- Develop students' skills in identifying and resolving common issues and bugs in web applications through systematic troubleshooting and problem-solving techniques.

1. Course Details

3.1 Theory (48 hrs)

Unit 1: Web Essentials (6 hrs)

The Internet, Basic Internet Protocols, the World Wide Web, HTTP (HTTP Request and Response Messages),FTP (Request, Response and Modes), Web Clients and Web Servers, Web Page and Web Site (Static and Dynamic),Web Application architecture(Tier architecture), Client Side and Server Side Scripting; Web 2.0, Web 3.0,Domain Registration, Web Hosting, Web application deployment, Using version control tools

Unit 2: HTML and CSS (12 hrs)

HTML Introduction, HTML Documents, HTML Elements and Attributes, Text Formatting Elements, Layout elements, Image elements, Table elements, Form elements, HTML 5 and its features, HTML 5 Elements and Attributes, XHTML and its feature, Introduction to CSS, CSS Syntax, Three Ways to apply CSS(Inline, Internal, External), CSS Selectors (Universal, ID, Class, Element, CSS combinators, Pseudo-classes, and Pseudo-elements, CSS Attributes selectors),Using CSS Units(Absolute and Relative),Layout related properties (Float, Display: inline, block, inline-block, flex, grid),Position properties (static, relative, fixed, absolute, sticky, z-index), Responsive Web Design, Introduction to Frontend Framework and Library

Unit 3: Client Side Scripting (12 hrs)

JavaScript: Introduction, Basic Syntax, Variables(let, const and var) and Data Types, Statements, Operators, Literals, Control Statements, Functions, Objects, Arrays, Built-in Objects, Strings, Forms and Regular Expression, Cookies, JavaScript Event Handling, Introduction HTML DOM, JavaScript Browser BOM, XML introduction, AJAX, JSON, ES6 and its Features (arrow function, spread operator, map, set, promises), Introduction to Client Side Libraries and Frameworks.

Unit 4: Server Side Scripting: PHP (12 hrs)

Introduction to server side scripting, Syntax, Variables, Data Types, Operators, Control Statements; Functions, Arrays, Strings, Superglobals, Forms, Form validations, Regular Expression for Server side validation, File Handling, Cookies, Session, Error and Exception, Database Handling(CRUD operation with DML queries), Introduction to Server Side Frameworks.

Unit 5: Web application security (6 hrs)

Credential stuffing, SQL Injection, Session Hijacking, Cross Site Scripting (XSS), Cross-site request forgery (CSRF), DDoS Attack, Using Client Side and Server Side technique to handle application attack, Using PDO for secure database transaction

3.2 Tutorial

Unit-specific exercises focused on problem-solving.

3.3 Laboratory Works

The practical work includes creating web pages and applications with using HTML, CSS, JavaScript, PHP, and other technologies mentioned in the syllabus. Students also learn to use client and server side frameworks for developing web applications.

- Learning to use version control tools.
- Design website for different application domain like; (Company, Portfolio, E-commerce and
- Making responsive design using media query
- Making interactive webpage using JS library and framework
- Setting up application using Client side framework
- Setting up application using Server side framework
- Implement server side script for backend CMS.
- Implement related security technique for securing web application
- Finally, student must host their project into web server and make live.

4. Evaluation Scheme

Evaluation of student's performances is divided into two main parts: (1) Internal Evaluation and (2) Semester examination, with breakdowns as follows:

Evaluation	Components	Weightage	Full Marks	Pass Marks
Internal Evaluation	Class Attendance and Performance	5	20	10
	Assignment	10		
	Seminar/Project/Presentation	5		
	Practical/Lab Examination	20	20	10
	Internal Examination	**20**	**20**	**10**
	Total Internal Marks		**60**	**30**
Semester Examination			**40**	**20**

5. Books

- Learning PHP, MySQL & JavaScript: a step-by-step guide to creating dynamic websites, Nixon - O'Reilly – 2021
- Jackson, J. C. (2007). Web Technologies: A Computer Science Perspective.
- Responsive Web design with HTML5 and CSS3 learn responsive design using HTML5 and CSS3 to adapt websites to any browser or screen size, Frain - Packt - 2012
- PHP and MySQL Web development, Welling & Thomson - Sams - 2002
- Web application security: exploitation and countermeasures for modern web applications Hoffman - O'Reilly - 2020

Data Structure and Algorithm

Course Code	BAI204	**Year/Semester**	II/I
Credit Weightage	3	**Lecture**	3 hrs/wk
		Tutorial	1 hrs/wk
		Practical	2 hrs/wk
		Total	96 hrs

1. Course Description

The Data Structures and Algorithms course is designed to provide students with a comprehensive understanding of fundamental data structures and algorithms that are essential for efficient and effective problem-solving in computer science and programming. This course forms the backbone of computer programming and software development, as it equips students with essential problem-solving skills and strategies to tackle real-world computational challenges.

2. Course Objectives

- Understand the fundamental concepts of data structures and algorithms, their significance, and their role in computer programming.
- Develop a strong foundation in problem-solving techniques and algorithmic analysis to efficiently solve complex computational problems.
- The course will cover the concept of recursion and how it can be employed to design elegant and efficient algorithms. Students will learn how to convert iterative solutions into recursive ones and vice versa.
- Learn different data structures and their applications, such as arrays, linked lists, stacks, queues, trees, graphs, and hash tables.
- Explore fundamental algorithms for sorting, searching, and graph traversal, and understand their time and space complexities.
- Study advanced algorithmic paradigms, such as divide-and-conquer, greedy algorithms, dynamic programming, and backtracking.

3. Course Details

3.1 Theory (48 hrs.)

Unit 1: Introduction to Data Structures & Algorithms (5 Hrs.)

Overview of algorithms and their significance in computer science. Characteristics of good algorithms: correctness, efficiency, and simplicity. Algorithm Design Techniques: Greedy algorithm, Divide and conquer algorithm, Dynamic programming, Backtracking and randomized algorithm. Algorithm analysis: time complexity, space complexity, and Big Oh notation.

Introduction to data structures, Abstract data type (ADT) and their role in organizing and storing data.

Unit 2: Arrays, Pointers and Linked Lists (5 Hrs.)

Introduction to arrays and their properties. Multi-dimensional arrays. Pointers and Dynamic Memory Allocation: malloc, calloc, realloc, free. Singly linked lists: structure, operations, and

applications. Doubly linked lists: structure, operations, and applications. Circular linked lists and their advantages.

Unit 3: Stack **(5 Hrs.)**

Basic Concept of Stack, Stack as an ADT, Stack Operations, Stack Applications, Stack implementation as Array and Linked List. Conversion from infix to postfix/prefix expression, Evaluation of postfix/ prefix expressions.

Unit 4: Queue **(5 Hrs.)**

Basic Concept of Queue, Queue as an ADT, Primitive Operations in Queue, Queue implementation as Array and Linked List. Linear Queue, Circular Queue, Priority Queue, Queue Applications. Double-ended queues (Deque) and their uses.

Unit 5: Recursion **(2 Hrs.)**

Principle of Recursion, Comparison between Recursion and Iteration. Example of recursive algorithms (factorial, Fibonacci Sequence, GCD, and Tower of Hanoi algorithms). Applications and Efficiency of Recursion.

Unit 6: Sorting **(8 Hrs.)**

Introduction; Internal and External Sorting; bubble sort, insertion sort, selection sort, merge sort, quick sort, shell sort, Heap Sort. Complexity of Sorting Algorithms.

Unit 7: Searching and Hashing **(5 Hrs.)**

Introduction to Searching, Search Algorithms: Sequential Search, Binary Search Efficiency of Search Algorithms.

Hashing: Hash Function and Hash Tables, Collision Resolution Techniques

Unit 8: Trees **(6 Hrs.)**

Concept and Definitions, Basic Operations in Binary Tree, Tree Height, Level and Depth, Binary Search Tree, Insertion, Deletion, Traversals, Search in BST, Balancing Tree: AVL tree, DSW Algorithm

Unit 9: Graphs **(7 Hrs.)**

Definition and Representation of Graphs, Graph Traversal, Minimum Spanning Trees: Kruskal and Prims Algorithm. Shortest Path Algorithms: Dijkstra's Algorithm

3.2 Tutorials

Unit-specific exercises focused on problem-solving.

3.3 Laboratory Works:

The laboratory work consists of implementing the algorithms and data structures studied in the course. Since all the units consists of their specified practicals except first unit, so students are expected to perform all the practical according to its unit lab work.

4. Evaluation Scheme

Evaluation of students' performance is divided into two main parts: (1) Internal evaluation and (2) Semester examination, with breakdown as follows.

Evaluation	Components	Weightage	Full Marks	Pass Marks
Internal Evaluation	Class Attendance and Performance	5	20	10
	Assignment	10		
	Seminar/Project/Presentation	5		
	Practical/Lab Examination	20	20	10
	Internal Examination	**20**	**20**	**10**
	Total Internal Marks		**60**	**30**
Semester Examination			**40**	**20**

5. Books

- Y Langsam, MJ Augenstein and A.M , Tanenbaum Data Structures using C and C++, Prentice Hall India, Second Edition 2015
- Leen Ammeral, Programmes and Data Structures in C, Wiley Professional Computing
- G.W Rowe, Introduction to Data Structure and Algorithms with C and C++ , Prentice Hall India
- Srivastava, S.K. & Srivastava, D. (2011). Data structure Through C in Depth, (2nd Ed.), BPB Publication

Microprocessor & Computer Architecture

Course Code	BAI205	**Year/Semester**	II/I
Credit Weightage	3	**Lecture**	3 hrs/wk
		Tutorial	1 hrs/wk
		Practical	2 hrs/wk
		Total	96 hrs

1. Course Description

This course is designed to provide students with a comprehensive understanding of microprocessor 8085 architecture, basic computer structure, central processing unit, microprogramed control, computer arithmetic, input/output, memory and pipelining mechanism.

2. Course Objectives

The course aims to achieve the following objectives:

- Introduce Microprocessor and Microprocessor based system.
- Explore features and functioning of intel 8085 microprocessor.
- Write an ALP using 8085 microprocessors.
- Familiarize basic computer.
- Identify the organization and design of basic computer.
- Explain microprogrammed control logic.
- Explore central processing unit.
- Familiarize with IO and memory organization.
- Identify computer arithmetic algorithms.
- Introduce pipelining mechanism.

3. Course Details

3.1 Theory (48 hrs.)

Unit 1: Introduction (4 hrs.)

Microprocessor, Components of Microprocessor: ALU, Register Array, Control Unit, Microprocessor as a CPU, Microprocessor based system, Microcontroller, Microprocessor vs Microcontroller, Von-Neumann and Harvard Architecture

Unit 2: Microprocessor 8085 (10 hrs.)

Features of 8085 MPU, Operations: Microprocessor-initiated, Internal and Peripheral operations. Pinout and Signals of 8085, Bus Organization, Flags, Interrupts, Register Organization, Functional block diagram, Instruction Set, Opcode and Operands, Addressing modes, Assembly Language Programming with 8085. Instruction Cycle and Timing diagrams, Machine Cycle & T-State.

Unit 3: Basic Computer Organization and Design (6 hrs.)

Basic Computer, Instruction codes, Computer Registers, Computer Instructions, Instruction Cycle, Memory-Reference Instructions, Input-Output and Interrupt, Design of Basic Computer, Design of Accumulator Logic.

Unit 4: Microprogrammed Control **(6 hrs.)**
Control Memory, Address Sequencing, Microprogram, Microinstruction format, Symbolic and Binary microprogram, Design of Control Unit.

Unit 5: Central Processing Unit **(6 hrs.)**
Register Organization, Stack Organization, Instruction Format, Addressing modes, Data Transfer and Manipulation, Program Control, RISC and CISC architecture.

Unit 6: Computer Arithmetic **(6 hrs.)**
Fixed-point vs Floating-point arithmetic, Fixed-point: Addition, Subtraction, Multiplication and Division algorithms.

Unit 7: Input/output and Memory Organization **(6 hrs.)**
Input/output: Peripheral Devices, Input-Output Interfaces, Modes of data transfer, Direct Memory Access (DMA), DMA Controller, Pin Diagram & Internal configuration of 8237 processor.
Memory: Memory Hierarchy, Auxiliary memory, Associative memory, Cache memory, Virtual Memory.

Unit 8: Pipelining **(4 hrs.)**
Pipelining, Arithmetic pipelining, Instruction pipelining, Pipelining Hazards.

3.2 List of Tutorials
The various tutorial activities that outfits this course should cover all the content of this course to give students a space to engage more actively with the course content in the presence of instructor. Students should submit tutorials as assignments or class works to the instructor for evaluation. The following tutorial activities of 15 hrs should be conducted to cover all the content of this course:

A. Discussion based Tutorials
- Comparative analysis of Von-Neumann, Harvard and modified Harvard Architecture.
- Applications of microprocessor and microcontroller based system in real world.
- Hardware and programmable module of 8 bit microprocessor.
- Interpret the concept of memory segmentation and pipelining in 16 bit microprocessor.
- Review each entity memory hierarchy for modern processors.
- Interfacing of DMA, interrupt controller and UART with 8085 microprocessor.
- Comparative analysis of different aspects of computing system as defined in Flynn's Classification.

B. Assembly level programming tutorials
- Assembly level programming illustrating data transfer instructions, arithmetic instructions, logical instructions and branching instructions in 8085 microprocessor.
- Enlighten PUSH and POP instructions using appropriate assembly level programs.
- Assembly level programming for simple sequencing program, multiplication, division, processing and sorting.
- Timing diagram for different types of instructions and RTL of each machine cycle involved during the execution of instructions.

C. Design Thinking tutorials
- Design a CPU for any given registers set, instruction set and state diagram. Show the RTL code for each execution cycle.
- Develop a control unit for any given state diagram.

- Design a micro-sequencer control unit for any given specifications following design procedure.

3.3 Laboratory Works

Laboratory works using 8085 Microprocessor Training Kit

- Illustrate Data Transfer, Arithmetic, Logical, Branching and Machine Control Instructions.
- Demonstrate instructions from each addressing mode.
- Perform 8 – bits addition, subtraction, multiplication and division.
- Perform 16 – bits addition and subtraction.
- Transfer block of data from one memory location to another.
- Arrange data in ascending/descending order.
- Find largest/smallest datum from data array.
- Identify even/odd, prime/composite number.
- Convert number from one base (radix) to another.

4. Evaluation Scheme

Evaluation of student's performances is divided into two main parts: (1) Internal Evaluation and (2) Semester examination, with breakdowns as follows:

Evaluation	**Components**	**Weightage**	**Full Marks**	**Pass Marks**
Internal Evaluation	Class Attendance and Performance	5	20	10
	Assignment	10		
	Seminar/Project/Presentation	5		
	Practical/Lab Examination	20	20	10
	Internal Examination	**20**	**20**	**10**
	Total Internal Marks		**60**	**30**
Semester Examination			**40**	**20**

5. Books:

- Ramesh Gaonkar, Microprocessor Architecture, Programming and Applications with the 8085, Penram International Publishing India
- M. Morris Mano, Computer System Architecture, PHI.
- Ajit Pal, Microprocessor: Principles and Applications, TATA McGraw-Hill
- William Stalling, Computer Organization and Architecture, Pearson

System Analysis and Design

Course Code	BAI206	**Year/Semester**	II/II
Credit Weightage	3	**Lecture**	3 hrs/wk
		Tutorial	1 hrs/wk
		Practical	2 hrs/wk
		Total	96 hrs

1. Course Description

This course is designed to provide students with a comprehensive understanding of System and System Development Ecosystem. The course covers different approaches of system analysis and design and various success and challenging real cases of System Development

2. Course Objectives

The course aims to achieve the following objectives:

- Introduce students to the fundamental concepts of System and System Design
- Equip the students with different quality issues of system.
- Familiarize students' system development based upon the valid customer's requirements.

3. Course Details

3.1 Theory (48 hrs)

Unit 1: Introduction to System Development (10 hrs)

Introduction of System Analysis and Design, Understanding the system, System Diagram, Development Process, Management Process, Supporting Processes, System Structure, ,Traditional waterfall SDLC , Other Approaches: Prototyping , Spiral, RAD, Introduction to Agile Development, CASE tools.

Unit 2: System Concept Formation (6 hrs)

Introduction to System Concept, Identifying and Selecting the System Development Projects, Corporate and Information Systems Planning, The Problem Identification, justifying a Solution, Generating broad alternative solution, Evaluating the proposal, Feasibility and its types, Selecting an alternative, Preparing statement of a User Requirement

Unit 3: Requirement Analysis (6 hrs)

Introduction to Requirement Analysis, Functional and non-functional requirement, User requirements, System requirements, Interface requirements, Information Collection, Questionnaires, Electronic Data Collection, Information by Observation, Using Ethnography, From observation to design, Information by Prototyping, Interface Prototyping. Prototyping Processes, Interviewing, Software requirement document.

Unit 4: Data Flow Diagram (DFD) (6 hrs)

Introduction to Data Flow Diagram, DFD symbols, Describing systems by DFD, Logical and Physical DFDS, Convention for good DFDS, DFDS Levelling.

Unit 5: Process Descriptions **(6 hrs)**

Introduction to Process Description, Process descriptions method, Structured English, Decision tables, Extended Decision tables, establishing logical correctness of Decision tables, Using Karnaugh maps to detect logical errors in decision tables, Eliminating redundant specifications.

Unit 6: Implementation and Maintenance **(8 Hrs.)**

System Implementation: Introduction, System Implementation, Software Application Testing, testing types Installation and its types Documenting the System, Training and Supporting Users, Organizational Issues in Systems Implementation; Maintaining Information Systems: Introduction, Maintaining Information Systems, Conducting Systems Maintenance

Unit 7: Object Modelling **(6 hrs)**

Introduction to Object Modelling, Object environment, Object structures, Modelling behaviour in object modelling, Unified Modelling Language (UML), Class and object diagram, Use case diagram, State diagram, Sequence diagram.

3.2 Tutorial

Unit-specific exercises focused on problem-solving.

3.3 Laboratory Work

In the practical session, students will learn to use project management, CASE, and modelling tools. They also prepare a project report that includes at least analysis, design, and implementation phases of system analysis and design.

4. Evaluation Scheme

Evaluation of student's performances is divided into two main parts: (1) Internal Evaluation and (2) Semester examination, with breakdowns as follows:

Evaluation	Components	Weightage	Full Marks	Pass Marks
Internal Evaluation	Class Attendance and Performance	5	20	10
	Assignment	10		
	Seminar/Project/Presentation	5		
	Practical/Lab Examination	20	20	10
	Internal Examination	**20**	**20**	**10**
	Total Internal Marks		**60**	**30**
Semester Examination			**40**	**20**

5. Books :

- Igor Hawryszkiewycz (IH), "System Analysis and Design"
- Jeffrey A. Hoffer, Joey F. George, Joseph S.Valacich (JH) "Modern System Analysis and Design"
- Ian Somerville (IS) "Software Engineering"